

NR2DT-03: Step 3 — Design

Series: NR2DT — New Relic to Dynatrace Migration Steps | **Notebook:** 3 of 10 |
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Overview

Goal of this step: lock in the target Dynatrace architecture before any artifact migrates. Bucket strategy, host group taxonomy, IAM model, and OpenPipeline routing are all decided here.

Procedural runbook — see **FAQ-01** (host group naming strategy) and **ORGNZ-02 / ORGNZ-99** (Grail bucket strategy) for the recommended design patterns this step uses.

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Prerequisites

Requirement	Details
Audience	Migration lead + assigned engineer for this step

Requirement	Details
Completed	NR2DT-02 — Strategize
Format	Procedural step — use as a runbook; defer to NRLC for depth
Topic deep dives	FAQ-01 (host group naming), ORGNZ-02 / ORGNZ-99 (bucket strategy), IAM-04/05 (policy design), IAM-11 (WORKSHOP)

1. What You'll Produce

Artifact	Format	Purpose
<code>bucket-design.md</code>	Markdown	Bucket names, retention, pricing model
<code>host-group-design.md</code>	Markdown	Host group taxonomy and naming
<code>iam-model.md</code>	Markdown	Groups, policies, scopes
<code>openpipeline-design.md</code>	Markdown	Enrichment + routing + filter rules
<code>terraform/</code>	HCL	Phase 1 Terraform module skeleton

2. Bucket Design

Use the recommended pattern from **ORGNZ-02 — Understanding Grail Buckets** (with the consolidated reference in **ORGNZ-99**) as your starting template:

Bucket	Retention	Pricing	Purpose
<code><org>_app_logs</code>	30 days	Retain w/ Included Queries	Application logs
<code><org>_infra_logs</code>	14 days	Retain w/ Included Queries	Infrastructure / K8s
<code><org>_security_logs</code>	365 days	Usage-based	Audit trail / PCI
<code><org>_spans</code>	14 days	Retain w/ Included Queries	Distributed traces
<code><org>_events</code>	90 days	Retain w/ Included Queries	Platform events
<code><org>_bizevents</code>	90 days	Retain w/ Included Queries	Business events

Use `<org>_` as the prefix — it lets IAM policies scope with one `storage:bucket-name` `startsWith "<org>_"` condition.

Document any deviations from this template (e.g., per-app buckets) and why.

3. Host Group Design

Per **FAQ-01 — Host Group Naming Strategy**, group hosts by ownership boundary, not by team alone. Common dimensions:

- Environment (`prod-app` , `nonprod-app` , `prod-infra`)
- Compliance scope (`payment-pci`)
- Workload tier (`prod-batch` , `prod-stream`)

Plan a host group per (environment × ownership) combination. Avoid a single all-hosts group — it breaks per-environment thresholds, IAM scoping, and automation safety.

4. IAM Model

Recommended starting groups:

Group	Bucket Access
<code><org>-admins</code>	All <code><org>_</code> buckets
<code><org>-sre</code>	All operational: app + infra logs, spans, events
<code><org>-app-eng</code>	App logs only
<code><org>-readonly</code>	All <code><org>_</code> buckets except security

Each statement grants a Grail table permission (`storage:logs:read` and so on) together with `storage:buckets:read` , scoped by the `storage:bucket-name` condition. The bucket permission is needed as well as the table permission — the IAM reference describes `storage:buckets:read` as *"Required additionally to a table permission."*

```
// <org>-admins
ALLOW storage:buckets:read, storage:logs:read, storage:spans:read,
storage:events:read, storage:bizevents:read
  WHERE storage:bucket-name startsWith "<org>_";

// <org>-sre
ALLOW storage:buckets:read, storage:logs:read WHERE storage:bucket-name IN
("<org>_app_logs", "<org>_infra_logs");
ALLOW storage:buckets:read, storage:spans:read WHERE storage:bucket-name =
"<org>_spans";
ALLOW storage:buckets:read, storage:events:read WHERE storage:bucket-name =
"<org>_events";

// <org>-app-eng
ALLOW storage:buckets:read, storage:logs:read WHERE storage:bucket-name =
"<org>_app_logs";

// <org>-readonly – security bucket excluded from the ALLOW scope
ALLOW storage:buckets:read, storage:logs:read, storage:spans:read,
storage:events:read, storage:bizevents:read
  WHERE storage:bucket-name startsWith "<org>_" AND storage:bucket-name
!= "<org>_security_logs";
```

Do not exclude the security bucket with a conditional DENY on a table permission.

It does not do what it reads as: *"Conditional DENY statements involving Grail table permissions are not supported and are executed as unconditional DENY."* A `DENY storage:logs:read WHERE storage:bucket-name = "<org>_security_logs";` bound to the SRE, app-eng or readonly group would remove **all** log access for that group. Exclude the bucket from the ALLOW scope instead, as `<org>-readonly` does above. If you do need a DENY, the only storage permission that accepts a conditional one is the bucket permission: *"Conditional DENY statements are supported only for the buckets:read permission."*

For policy design in depth (parameterized policies, boundaries), see **IAM-04** and **ORGNZ**.

Sources: [IAM policy statement syntax \(DT docs\)](#), [IAM policy reference \(DT docs\)](#), read 10/05/2026.

5. OpenPipeline Layout

Routing rules in priority order (first match wins):

Priority	Matcher	Bucket
1	<code>matchesValue(log.source, "*audit*") OR loglevel == "CRITICAL"</code>	<code><org>_security_logs</code>
2	<code>in(k8s.namespace.name, {"kube-system", "kube-public", ...})</code>	<code><org>_infra_logs</code>
3	App namespace match	<code><org>_app_logs</code>
4	catch-all	<code>default_logs</code> (review weekly)

Enrichment rules for any NR workload identifier (replaces NR's tag/workload model):

```
matcher: contains(k8s.namespace.name, "prod")
field: environment
value: prod
```

Drop rules for cost optimization (decide which apply now vs. Wave 5):

- Drop health-check logs (`contains(content, "GET /health")`)
- Drop DEBUG logs in production
- Sample sidecar proxy logs

6. Step Exit Criteria

S3 — Architecture Locked

- `[] bucket-design.md` final; bucket names + retention + pricing model documented
- `[] host-group-design.md` final; taxonomy understood by SRE + service owners
- `[] iam-model.md` final; security stakeholder signed off
- `[] openpipeline-design.md` final; routing rules sequenced
- `[]` Phase 1 Terraform module skeleton committed (will be applied in Wave 0 of Step 5)

Next step: NR2DT-04 — Translate (NRQL → DQL run; manual triage).

This notebook was AI-generated from community-submitted and publicly available sources, including the open-source [NewRelic-to-Dynatrace-Migration-Utilities](#), [nrql-engine](#), and [nrql-translator](#) projects. This

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